

dbt Certification ⚡️⚡️⚡️ Checkpoint 2: Govern and Debug Your Models

Checkpoint 2 — Govern and Debug Your Models (Meta Study Guide)

Synthesized exam-prep guide for Checkpoint 2. Reorganizes the 7 per-resource cheat sheets by topic and shows how they fit together as a single story: **how to publish reliable model APIs and debug when things go wrong**.

The shape of Checkpoint 2

CP2 is significantly tighter than CP1 — 7 resources, two themes:

- Governance** — contracts, versions, constraints, data-product management. How to make a model into a trustworthy API for downstream consumers.
- Debug & control** — compile, debugging errors, behavior changes, global configs. How to inspect what dbt is doing and configure its behavior.

⚠️ The Checkpoint 2 readings list one item, *Data product management: best practices*, hosted at www.getdbt.com/blog/data-product-management. That post lives on dbt's marketing site, which is **not open-sourced** and was unreachable from this sandbox. I've flagged it here; if you can paste the contents, I can produce a per-resource cheat sheet for it. The synthesis below incorporates its themes based on the rest of the CP2 material and the Mesh courses it draws from.

1. The “models as APIs” mental model

Read: 01-model-contracts.md, 02-model-versions.md, 03-constraints.md.

The core idea: a public dbt model has **producers** (your team) and **consumers** (other teams, dashboards, ML pipelines, other dbt projects). Producers want freedom to iterate; consumers want stability. CP2 is the toolkit that makes both possible:

Concern	Tool
What columns/types will the model produce?	Contract
How do I change those guarantees without breaking consumers?	Versions + deprecation_date
What guarantees does the warehouse enforce?	Constraints
How do producers and consumers coordinate at scale?	Data product management practices (the missing post)

This is conceptually the same arc you see for web APIs — define the contract, version it, deprecate old versions on a schedule, communicate clearly.

2. Contracts — shape guarantees enforced at build time

Read: 01-model-contracts.md.

A contract defines a model's column names, data types, and (optionally) constraints. With `enforced: true`:

- Preflight check** — dbt verifies the SQL *would* produce columns matching the contract before sending DDL.
- DDL enforcement** — column types and constraints land in the `CREATE TABLE` statement; the warehouse enforces them on insert.

```
models:
  - name: dim_customers
    config:
      materialized: table # required: table or incremental, NOT view/ephemeral
      contract: {enforced: true}
    columns:
      - name: customer_id
        data_type: int
        constraints:
          - type: not_null
      - name: customer_name
        data_type: string
```

Hard rules

- Materialization must be `table` or `incremental`.
- Every** column needs `data_type`.
- Ephemeral and view models can't carry contracts.

Contracts vs data tests

	Contract	Data test
Validates	shape (columns/types)	content (rows)
Failure effect	Build doesn't happen	Test reports failure
When	Build-time	After build
Configurable severity	No	Yes

For exam purposes: contracts catch *structural* breakage; tests catch *data quality* issues. Some platform-enforced constraints (`not null` everywhere; everything else is platform-dependent) can replace equivalent data tests — cheaper, build-time enforcement.

Which models should have contracts?

Public models — anything other teams, projects (Mesh), or external systems consume. Internal intermediate models usually don't need them.

3. Versions — evolve models without breaking consumers

Read: 02-model-versions.md.

When a contract change would break consumers (drop column, rename, type change), you don't just push it. You **create a new version**:

```
models:
  - name: dim_customers
    latest_version: 1
    config:
      materialized: table
      contract: {enforced: true}
    columns:
      - name: customer_id
        data_type: int
      - name: country_name
        data_type: varchar
    versions:
      - v: 1 # current latest; matches the shared spec
      - v: 2 # the new shape
        columns:
          - include: all
            exclude: [country_name] # breaking change: dropping country_name
```

How ref resolves

- `ref('dim_customers')` → resolves to **latest** automatically.
- `ref('dim_customers', v=1)` → pins to v1.
- dbt prints notices when an unpinned ref resolves to latest *and* a prerelease exists.

File / relation conventions

v	role	file name	relation
3	prerelease	dim_customers_v3.sql	dim_customers_v3
2	latest	dim_customers.sql or dim_customers_v2.sql	dim_customers_v2 AND dim_customers (recommended)
1	old	dim_customers_v1.sql	dim_customers_v1

The deprecation loop

- Develop new version.
- Bump latest to point at it.
- Mark old for deprecation (`deprecation_date` resource property).
- Wait for consumers to migrate.
- Delete old version.

Tactical notes

- Only version **public/contracted models** with downstream commitments. Don't version every change.
- Keep ≤ 2–3 live versions at a time (same rule as web APIs).
- Non-breaking additions (new column, bug fix) don't need a new version — update latest in place.
- Schedule periodic version bumps (1–2× per year) where you sweep out unused columns rather than versioning every small change.

Selecting by version

```
selectors:
  - name: exclude_old_versions
    default: "{{ target.name == 'dev' }}"
    definition:
      method: fqqn
      value: "*"
      exclude:
        - method: version
          value: old
```

Useful for skipping deprecated versions in dev while still building them in prod during migration.

4. Constraints — warehouse-enforced rules

Read: 03-constraints.md.

Constraints attach to columns or models, declaring rules the warehouse enforces on insert. Types: `not_null`, `unique`, `primary_key`, `foreign_key`, `check`, `custom`.

Prerequisites

- Materialization: `table` or `incremental`.
- The model has an **enforced contract**.
- Every column declares `data_type`.

Column-level vs model-level

- ✓ Single-column constraints → **column-level**.
- ✓ Multi-column constraints (composite PK, multi-column FK, cross-column check) → **model-level**.
- Multiple `primary_key` constraints must be model-level.

```
models:
  - name: dim_customers
    config: {materialized: table, contract: {enforced: true}}
    constraints:
      - type: primary_key
        columns: [customer_id, region]
      - type: foreign_key
        columns: [customer_id]
        to: ref('stg_customers')
        to_columns: [customer_id]
      - type: check
        columns: [start_date, end_date]
        expression: "start_date <= end_date"
        name: dim_customers_valid_window
    columns:
      - name: customer_id
        data_type: int
        constraints: [{type: not_null}]
```

Platform enforcement reality

- **Postgres** enforces everything.
- **Most analytical platforms** enforce only `not_null`; PK/UQ/FK are *informational*.
- Use `warn_unenforced: false` to silence noise on supported-but-not-enforced constraints; `warn_unsupported: false` for those the platform doesn't recognize.

Foreign keys (v1.9+)

```
- type: foreign_key
  columns: [customer_id]
  to: ref('stg_customers')
  to_columns: [customer_id]
```

Using `ref()` for `to:` captures the dependency in the DAG and survives environment switches — much better than hardcoding schema names in `expression`.

Constraints vs data tests

Constraints win when the platform actually enforces them — they fail the build at insert time, no extra query. For everything informational, pair with a data test.

5. Data product management (the missing-blog gap)

The Checkpoint 2 readings list one resource I couldn't fetch — *Data product management: best practices* on www.getdbt.com. I'll note its themes here based on adjacent dbt material; replace this section if you can pull the actual post.

Why this complements the technical CP2 toolkit

Contracts, versions, and constraints give you the *mechanics* of treating models as APIs. The data-product-management practices give you the *organizational* side:

- **Producers** (data engineers / analytics engineers) are responsible for the contract, freshness SLA, and migration plan. They publish a versioned, documented product.
- **Consumers** (BI teams, ML engineers, downstream dbt projects via Mesh) are responsible for staying within the contract and migrating when notified.
- **The contract** is the formal boundary; everything else (tests, exposures, freshness, version-bump cadence) is the SLA.

This is the same idea as product management for software APIs: ownership, lifecycle, communication, deprecation.

How CP2 features support this division

Practice	Mechanism in dbt
"Declare the API"	Contract
"Communicate the API"	Description + doc blocks; rendered docs site
"Make consumers visible"	Exposures (CP3)
"Version & sunset"	Versions + <code>deprecation_date</code>
"Health monitoring"	Source freshness, data tests, data health tiles
"Restrict access"	Groups + <code>access: public/private/protected</code> (CP1's rest-of-project + Mesh)

Action items the missing post almost certainly recommends

- Treat each public model as a product with an owner, an SLA, and consumers.
- Bake the migration cadence (e.g., once-or-twice-yearly version bump) into the team's calendar.
- Use exposures and Catalog/Explorer to keep producers and consumers visible to each other.
- Tie test severity to consumer impact (this is also the *Test smarter* framework in CP3).

6. Debugging — the systematic approach

Read: 04-debugging-errors.md.

The five-step loop

1. **Read the error message** — names the type and the file.
2. **Open the file** — fix is often obvious.
3. **Isolate** — run one model; revert recent changes.
4. **Use compiled files and logs:**
 - `target/compiled/<model>.sql` — bare `select`, runnable in a SQL client.
 - `target/run/<model>.sql` — the SQL dbt actually sent (with materialization DDL).
 - `logs/dbt.log` — every query dbt ran. Recent errors at the bottom.
5. **Ask for help** — write a good question, with the error + the file content.

Error type → likely cause

Type	Phase	Likely cause
Runtime Error	Initialize	Not a dbt project; bad profile; failed connection; invalid <code>dbt_project.yml</code>
Compilation Error	Parsing	Invalid Jinja; invalid YAML; bad <code>ref</code> ; wrong schema key
Dependency Error	Graph validation	Circular <code>ref</code> ; disabled model still <code>ref'd</code>
Database Error	SQL execution	Type mismatch; missing column; permissions; dialect mismatch

Tools that solve most debugging

- `dbt debug` — connection check.
- `dbt debug --config-dir` — find your `profiles.yml`.
- `dbt parse` — fast syntax check, no warehouse.
- `dbt compile` — render SQL; inspect what dbt is sending.
- `dbt show --select <test>` — see the failing rows for a test.
- `--debug` flag — verbose log output.
- `--fail-fast` — stop at first error.

Diagnostic pro tip

When a database error shows up in a downstream model but feels off, temporarily materialize the upstream chain as tables. The warehouse will throw the error at the real culprit.

7. dbt compile

Read: 05-cmd-compile.md.

Renders compiled SQL into `target/` without executing it. Used to: - Inspect rendered Jinja/macros/refs. - Manually run a model's SQL in a SQL client for debugging. - Compile analyses (analyses aren't materialized any other way).

Common patterns

```
dbt compile                                # everything
dbt compile --select stg_orders            # one model
dbt compile --inline "select * from {{ ref('raw_orders') }}" # ad-hoc
dbt compile --no-introspect               # avoid metadata queries

dbt --no-populate-cache compile           # skip warehouse cache warm-up
```

compile vs parse vs run vs build

Command	Compiles	Connects to warehouse	Executes
dbt parse	✗	✗	✗
dbt compile	✓	usually (cache/introspection)	✗
dbt run	✓	✓	✓ models
dbt build	✓	✓	✓ everything

v1.12+ extends `compile` to **snapshots** — you can finally inspect the generated SCD SQL.

8. Behavior changes — controlled migration of dbt's runtime

Read: 06-behavior-changes.md.

A specific category of flags that gate **breaking dbt-runtime changes**. They live in `dbt_project.yml`'s `flags:` block, version-controlled and PR-reviewed.

Three-phase lifecycle

1. **Introduction** — new behavior available but disabled by default.
2. **Maturity** — default flips to `true`; old behavior still selectable but warned about.
3. **Removal** — flag and old behavior deleted from dbt.

What counts as a behavior change

- Same code + same command → different result.
- New errors (not warnings).
- Macro signature changes.
- Breaking changes to artifact JSON.

What's NOT a behavior change (handled in regular releases)

- Bug fixes.
- New warnings.
- New artifact fields with defaults.

The flags you'll see most often

```
flags:
  require_explicit_package_overrides_for_builtin_materializations: true

  require_resource_names_without_spaces: true
  source_freshness_run_project_hooks: true
  state_modified_compare_more_unrendered_values:
false # turn ON to reduce CI false positives
  validate_macro_args: false
  require_generic_test_arguments_property: true
```

Fusion erases the migration

All behavior-change flags are removed in Fusion — the new behavior is always on.

9. Global configs — how dbt runs

Read: 07-global-configs.md.

Flags vs resource configs: - **Resource configs** describe **what** to build (materialization, tags, schema). - **Flags / global configs** describe **how** dbt runs (logging, fail-fast, threads, partial parse, defer, target).

Three layers of setting (most specific wins)

- 1. CLI option (--full-refresh).
- 2. Environment variable (DBT_F00 v1.10-; DBT_ENGINE_F00 v1.11+).
- 3. dbt_project.yml under flags:.

Common flags

- Logging: debug, log_level, log_format, log_path, quiet, printer_width, use_colors.
- Behavior: fail_fast, warn_error, warn_error_options, partial_parse, populate_cache.
- Targets: target, profile, profiles_dir, project_dir.
- Slim CI: defer, defer_state, favor_state, state.
- Build cost: empty, sample, event_time_start, event_time_end.
- Resource type: resource-type, --exclude-resource-type.
- Failures: store_failures.

Reading flags from Jinja

```
on-run-start:
  - '{{ log("I will fail fast", info=true) if flags.FAIL_FAST }}'
```

Don't use flags for parse-time-resolved configs (refs, sources).

10. Putting CP2 together

Imagine you own a public dim_customers mart consumed by three dashboards and another dbt project via Mesh.

- 1. **Contract.** Declare every column with its data_type. Add not_null and primary_key constraints on customer_id. Materialize as table.
- 2. **Tests.** PK uniqueness + not-null (data tests); business-anomaly tests on the calculated fields (CP3 Test smarter).
- 3. **Exposures.** Declare the three dashboards as exposures so the lineage is explicit (CP3).
- 4. **Version.** Today the model is v1. Tomorrow you need to drop country_name. Create dim_customers_v2.sql, declare versions: in YAML, set latest_version: 2, give v1 a deprecation_date. Inform consumers.
- 5. **Debug.** A consumer reports unexpected NULLs. You dbt show --select unique_dim_customers_customer_id to see failing rows; trace to the upstream staging model. Fix and dbt build --select stg_customers+.
- 6. **Behavior changes.** Upgrading to next dbt version surfaces a warning about state:modified comparisons. Turn on state_modified_compare_more_unrendered_values to reduce CI false positives.
- 7. **Compile.** Add a complex check constraint expression; dbt compile --select dim_customers to confirm the DDL renders correctly before running.

Exam-ready summary tables

Governance tool → role

Tool	Role
Contract	Shape guarantee (columns/types) enforced before/during build
Version	Mechanism for evolving contracts without breaking consumers
Constraints	Per-column / per-model rules the warehouse enforces
deprecation_date	Sunset signal for old versions
Exposures (CP3)	Make consumers visible

Materialization compatibility

	Contract	Constraints (enforced)
view	✗	✗
ephemeral	✗	✗
table	✓	✓
incremental	✓	✓
materialized_view	varies	varies

Behavior-change-flag lifecycle phases

- 1. Introduction (default false).
- 2. Maturity (default flips to true).
- 3. Removal (flag and old behavior deleted).
- 4. Fusion: all flags removed.

Compile-related commands

Command	When
dbt parse	Validate parse without warehouse
dbt compile	Render compiled SQL into target/
dbt compile --inline	One-off Jinja-SQL rendering
dbt show	Compile + execute + preview rows

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How to study from here

- 1. Walk through the “publishing a public model” scenario in \$10 mentally.
- 2. Memorize the contract prerequisites and the version naming conventions.
- 3. Know not_null is the only constraint most analytical warehouses enforce.
- 4. Reproduce the four error-type → likely-cause table.
- 5. Distinguish flags: from resource configs cold.
- 6. Move to Checkpoint 3.

Model Contracts

Source: https://docs.getdbt.com/docs/mesh/govern/model-contracts Type: Documentation · Checkpoint 2

What it is

A model contract is a set of upfront guarantees about a model's **shape** — its columns, their data types, and (optionally) constraints. With a contract enforced, dbt verifies before building that the model's SQL will produce a dataset matching those guarantees. If it doesn't, the build fails.

Why use one

A bare select model can change shape whenever its SQL changes. That's great for iteration, but bad for downstream consumers (dashboards, other dbt projects, external systems). Contracts protect those consumers by making the model's interface explicit and validated.

Defining a contract

```
# models/marts/customers.yml
models:
  - name: dim_customers
    config:
      contract:
        enforced: true
    columns:
      - name: customer_id
        data_type: int
        constraints:
          - type: not_null
      - name: customer_name
        data_type: string
```

When enforced: 1. **Preflight check** — dbt verifies the SQL will return columns with the declared names and data_types. Column order doesn't matter at this stage. 2. **DDL inclusion** — dbt includes the column names, types, and constraints in the

CREATE TABLE DDL it sends to the warehouse. The warehouse enforces them on insert. The DDL also orders columns to match the contract.

Requirements

- Every column must declare name AND data_type.
- Constraints (beyond just types) are **only** valid on table or incremental materializations — not on view or ephemeral.
- Constraints depend on platform support (see Constraints doc).

Contracts vs tests

	Model contract	Data test
Validates	Shape (columns/types)	Content (rows)
When	Build-time (preflight + DDL)	After build
Effect on build	Failed contract → no build	Failed test → table built, test reports
Configurable severity	No	Yes (severity: warn etc.)
Best for	“API” guarantee for downstream	Data quality

Some data tests can be replaced by **constraints** for build-time enforcement (cheaper, stricter): - Prerequisites: table or incremental materialization, full contract, platform actually enforces the constraint (most enforce only not_null). - Most platforms accept primary_key/unique/check as informational but don't enforce them at insert time.

Which models should have contracts?

Recommended for **public models** that other groups/teams/projects consume — internally via dbt Mesh or externally via dashboards/exposures. For private intermediate models, contracts are overkill.

All columns are required

A contract must cover **every** column the model selects. There's no partial-contract option. For wide models this means a lot of YAML — generation tools (dbt-codegen) help.

Breaking changes

When you change a contracted model in a way that violates the contract (drop a column, change a type), dbt detects it during state comparison: - **Versioned models** → error (breaking change requires bumping version). - **Unversioned models** → warning. - Removing a contracted model entirely (delete/rename/disable) → same rules apply (v1.9+).

This is how contracts pair with **model versions** — contracts make the breaking change detectable; versions give you a path through it.

Platform constraint support (high level)

- Postgres: full ANSI + check (all enforced).

Model Versions

Source: <https://docs.getdbt.com/docs/mesh/govern/model-versions> **Type:** Documentation · Checkpoint 2

What it is

A mechanism for shipping multiple **live, simultaneous versions** of the same model so producers can iterate on breaking changes while consumers get a migration window. Versioning is to dbt models what semver is to public APIs.

The producer/consumer tension

- **Producer** wants to evolve logic and structure.
- **Consumer** wants stability and notice before things break.

Model versions resolve this by letting both happen concurrently for a while.

When to version

- A change is **breaking**: column dropped, renamed, type changed, nullability flipped — anything that breaks a contracted shape.
- A recalc that doesn't break the contract but materially changes results (judgment call).

Don't version every change. Non-breaking additions (new column, bug fix in an existing column's logic) should go on the latest version. Plan periodic version bumps (1–2x a year) where you sweep out unused columns.

Versions vs version control vs new model

	Version control	New model	Model versions
Lives in repo?	Always 1 state on main	Independent model	Multiple live versions
Deployed at once?	One environment at a time	One name	Multiple at once
Notifies consumers?	No	No	Yes (prerelease/deprecation)
Reusable config?	–	No	Yes (define diffs only)

How ref resolves

- ref('dim_customers') → resolves to the model's **latest** version automatically.
- ref('dim_customers', v=1) → pins to v1.

dbt prints helpful messages when an unpinned ref resolves to latest and a prerelease is available:

Found an unpinned reference to versioned model 'dim_customers'.
Resolving to latest version: my_model.v2
A prerelease version 3 is available...

File / relation naming convention

v	version	ref syntax	File name	Relation
3	prerelease	ref('dim_customers', v=3)	dim_customers_v3.sql	analytics.dim_customers_v3
2	latest	ref('dim_customers') or ref(..., v=2)	dim_customers_v2.sql or dim_customers.sql	analytics.dim_customers_v2 and analytics.dim_customers (recommended)
1	old	ref('dim_customers', v=1)	dim_customers_v1.sql	analytics.dim_customers_v1

- Most analytical platforms: enforce not_null only; others are informational.
- Use warn_unenforced: false and warn_unsupported: false per-constraint to silence warnings.

Key takeaways

- Contracts = shape guarantees enforced before and during build.
- enforced: true + name + data_type for every column.
- Constraints work only on table / incremental materializations.
- Distinct from tests: contracts validate shape, tests validate rows.
- Pair with **model versions** to manage breaking changes gracefully.
- Recommend for public/consumed models, not every internal one.

Related

- Constraints (the per-constraint reference).
- Model versions (how to evolve contracts safely).
- Exposures (mark downstream consumers).

Defining versions in YAML

Diffs-only style (recommended)

```
models:
  - name: dim_customers
    latest_version: 1
    config:
      materialized: table
      contract: {enforced: true}
    columns:
      - name: customer_id
        data_type: int
      - name: country_name
        data_type: varchar
    versions:
      - v: 1
        # matches what's above
      - v: 2
        # breaking change: dropped country_name
        columns:
          - include: all
            exclude: [country_name]
```

Fully specified

You can also re-state config and columns inline under each version entry.

include / exclude helpers

include: all plus exclude: [col1, col2] produces the diff cleanly — useful when wide models change only a few columns.

Deprecation

Pair versions with deprecation_date to announce a sunset date for old versions. dbt warns consumers as the date approaches. Once past it, the version should be deleted from the project.

The whole loop: **develop new version** → **bump latest** → **mark old for deprecation** → **wait for migration** → **delete old**.

Selection by version

dbt- selectors support method: version:

```
selectors:
  - name: exclude_old_versions
    default: "{ target.name == 'dev' }"
    definition:
      method: fqfn
      value: "*"
      exclude:
        - method: version
          value: old
```

Lets you skip old versions in dev while still building them in prod during the migration window.

Practical guidance

- Version only models with downstream commitments (public/contracted models).
- Avoid having more than 2–3 live versions at a time — same wisdom as web APIs.
- Communicate deprecation dates clearly and well in advance.
- Each version has its own warehouse relation — be aware of storage cost.

Key takeaways

- Model versions = multiple live versions of one logical model.
- Breaking changes → new version; non-breaking → just update latest.
- ref(name, v=N) to pin; ref(name) for latest.
- Define in YAML with versions: + latest_version + per-version diffs.
- Pair with contract to detect breaks and deprecation_date to sunset.

Related

- Model contracts (catches the breaking change).
- deprecation_date resource property.
- ref with version argument.
- state:modified.body / version-aware state selection.

Constraints

Source: <https://docs.getdbt.com/reference/resource-properties/constraints> **Type:** Documentation · Checkpoint 2

What they are

Platform-enforced data integrity rules attached to a model's columns or the model itself. When supported and enforced by the warehouse, they reject invalid data at insert/load time — failing the build cleanly rather than silently writing bad rows.

Prerequisites

- Materialization is table or incremental — **never** view or ephemeral.
- The model has an enforced **contract** (contract: {enforced: true}).
- Every column has `data_type` declared (consequence of the contract requirement).

Constraint types

`not_null`, `unique`, `primary_key`, `foreign_key`, `check`, `custom`.

Structure

```
constraints:
- type: <type>           # required
  expression: "...\"      # required for check; optional otherwise
  name: \"...\"             # optional, platform-dependent
  columns: [c1,
c2] # model-level only – list of cols the constraint covers
    # v1.9+ foreign-key inputs
    to: ref('other_model') # or source('src','tbl')
    to_columns: [other_col]
    # warning controls
    warn_unenforced: false # supported but not enforced (e.g. PK on
Snowflake)
    warn_unsupported: false # not supported by platform (e.g. check on
Redshift)
```

Column-level vs model-level

-  **Column-level:** prefer this for single-column constraints.
-  **Model-level:** required for *multi-column* constraints (e.g., composite primary keys, multi-column FKs, cross-column `check`).
- **Multiple `primary_key`** constraints must be defined at the model level. Column-level multi-PK is **not** supported.

```
models:
- name: dim_customers
  config:
    materialized: table
    contract: {enforced: true}
  constraints:
  - type: primary_key
    columns: [customer_id, region]
  - type: foreign_key
    columns: [customer_id]
    to: ref('stg_customers')
    to_columns: [customer_id]
  - type: check
    columns: [start_date, end_date]
    expression: "start_date <= end_date"
    name: dim_customers_valid_window
  columns:
  - name: customer_id
    data_type: int
    constraints:
    - type: not_null
  - name: status
    data_type: varchar
    constraints:
    - type: check
      expression: "status in ('active', 'inactive')"
```

Foreign keys (v1.9+)

The `to + to_columns` form uses `ref()/source()` — that means: - dbt captures the dependency in the DAG. - The reference works across dev/CI/prod (no hardcoded schemas).

Debugging Errors

Source: <https://docs.getdbt.com/guides/debug-errors> **Type:** Documentation · Checkpoint 2

The general debugging loop

1. **Read the error message.** dbt error messages name the error *type* and the *file* where it happened.
2. **Open the offending file.** Often the fix is obvious.
3. **Isolate.** Run one model at a time; revert the change that broke things.
4. **Use compiled files and logs.**
 - `target/compiled/` — select statements you can run in any SQL editor.
 - `target/run/` — the SQL dbt executes (with materialization wrappers).
 - `logs/dbt.log` — all queries dbt ran, plus structured logging. **Recent errors at the bottom.**
 - dbt Cloud: use the `Details` tab in command output.
5. **Ask for help** — but write a good question first.

Pre-1.9 / unsupported adapters: fall back to a free-text `expression`.

Platform support overview

Platform	not_null	unique / PK	FK	check
Postgres	enforced	enforced	enforced	enforced
Redshift	enforced	informational	informational	not supported
Snowflake	enforced	informational	informational	informational
BigQuery	enforced	informational	informational	not supported
Databricks/Spark	enforced	informational	informational	varies

(See Model contracts doc for the full platform matrix.)

What “unenforced” means

The constraint is included in DDL metadata so external tools (catalogs, ER diagram tools) see it — but the warehouse will happily insert violating rows. For these, pair with a `unique/not_null` **data test** to actually catch bad data.

Constraints vs data tests

- **Constraints** validate at insert time. Cheap (no extra query) but limited (most platforms enforce only `not_null`).
- **Data tests** validate after build. Flexible, configurable severity, can run anywhere — and you can `store_failures` for inspection.

Use constraints where they’re **enforced** and you need build-time stops. Use data tests where you need flexibility or the platform doesn’t enforce.

Postgres example DDL produced

```
create table "db"."schema"."constraints_example_dbt_tmp" (
  id integer not null primary key check (id > 0),
  customer_name text,
  first_transaction_date date
);
```

Common patterns

```
# Composite PK
constraints:
- type: primary_key
  columns: [order_id, line_item_id]

# FK to another model
- type: foreign_key
  columns: [customer_id]
  to: ref('dim_customers')
  to_columns: [customer_id]

# Cross-column check
- type: check
  expression: "shipped_at >= ordered_at"
  columns: [shipped_at, ordered_at]

# Silence warnings on unenforced
- type: primary_key
  columns: [id]
  warn_unenforced: false
```

Key takeaways

- Constraints require an enforced contract on a `table` or `incremental` model.
- Six types: `not_null`, `unique`, `primary_key`, `foreign_key`, `check`, `custom`.
- Column-level for single columns; model-level for multi-column or multiple PKs.
- Most analytical warehouses only enforce `not_null` — others are informational.
- Use `warn_unenforced/warn_unsupported` to silence noise.
- Pair informational constraints with data tests to actually catch violations.

Related

- Model contracts (precondition).
- Data tests, custom generic tests, `store_failures`.
- Adapter resource configs for per-platform details.

Error types (by execution phase)

Phase	What dbt is doing	Error type
Initialize	Confirm this is a dbt project + can reach the warehouse	Runtime Error
Parsing	Validate Jinja in <code>.sql</code> , YAML in <code>.yaml</code>	Compilation Error
Graph validation	Build dependency graph, check it’s acyclic	Dependency Error
SQL execution	Run the models	Database Error

Runtime Errors (initialization)

“Not a dbt project”

Missing `dbt_project.yml` in CWD or parents. - `pwd` to check directory. - `ls` to confirm `dbt_project.yml` is there.

“Could not find profile”

The `profile:` key in `dbt_project.yml` doesn’t match any block in `profiles.yml`. Usually a typo (singular vs plural). Run `dbt debug --config-dir` to locate `profiles.yml`.

“Failed to connect”

Credentials wrong. Update `profiles.yml`, then `dbt debug` to test:

```
dbt debug
# Connection test: OK connection ok
```

Invalid `dbt_project.yml`

"Additional properties are not allowed ('hello' was unexpected)" — you have a key `dbt` doesn't recognize. Either remove/rename it, or check `dbt --version` (key may belong to a newer release).

Compilation Errors (parsing)

Invalid `ref`

"depends on a node named 'stg_customer' which was not found" — typo on a model name. Open the SQL file, fix the `ref()` argument, or rename the upstream file.

Invalid Jinja

"Reached EOF without finding a close tag" — missing `{% endmacro %}`, `{% endif %}`, `}`, etc. Read the line number in the error.

Invalid YAML

Indentation error → the YAML parser can't build a dict. Use: - Indentation guides in your editor. - A YAML validator (`yamllint`, `yamllint.com`).

Incorrect YAML spec

YAML is well-formed but uses a key `dbt` doesn't recognize for that resource. Check the reference for the resource type.

Dependency Errors (graph)

- **Circular dependency** — Model A refs Model B which refs Model A. Break the cycle (often by pulling a shared concept up into an intermediate).
- **Disabled model still referenced** — A disabled model can't be `ref`'d. Either enable it or remove the reference.

Database Errors (execution)

The warehouse rejected the SQL `dbt` sent. - Read the warehouse-specific error in the `dbt error` message. - Open `target/run/<model>.sql` (the exact SQL `dbt` sent) and try it in your warehouse client. - Common causes: type mismatch, missing column, permissions, syntax dialect difference.

dbt compile

Source: <https://docs.getdbt.com/reference/commands/compile> **Type:** Documentation · Checkpoint 2

What it does

Generates executable SQL for models, data tests, analyses, functions (and, in v1.12+, snapshots) **without executing it against the warehouse**. The output lands in `target/`.

What you can compile

- Models (v1.7+)
- Data tests
- Analyses
- Functions (UDFs)
- Snapshots (v1.12+)

Why use it

- **Inspect compiled SQL** — verify Jinja, macros, and refs resolved correctly.
- **Manually run a model's SQL** — copy the compiled `select` and execute in your warehouse client to debug a database error.
- **Render analyses** — the only way to materialize their SQL (analyses aren't built by `dbt run`).

Common misconceptions

-  `dbt compile` is NOT required before `dbt run/dbt build/dbt test`. Those commands compile internally.
-  Use `dbt parse` (not `compile`) if you only want to validate that the project loads, no warehouse connection involved.

Interactive compile (v1.5+)

Print compiled SQL directly to the terminal — useful for one-offs.

--select a single named node

```
dbt compile --select stg_orders
```

Output:

```
Compiled node 'stg_orders' is:
with source as (
  select * from "jaffle_shop"."main"."raw_orders"
), renamed as (
  select id as order_id, user_id as customer_id, order_date, status
  from source
)
select * from renamed
```

--inline an arbitrary dbt-SQL query

```
dbt compile --inline "select * from {{ ref('raw_orders') }}"
```

Output:

Useful commands for debugging

Command	What it tells you
<code>dbt debug</code>	Project setup + connection test
<code>dbt debug --config-dir</code>	Where your <code>profiles.yml</code> lives
<code>dbt parse</code>	Parse without executing — finds syntax errors fast
<code>dbt compile --select my_model</code>	Renders the SQL to <code>target/compiled/</code>
<code>dbt compile --inline "..."</code>	Compile arbitrary Jinja-SQL inline
<code>dbt show --select unique_my_model_id</code>	Preview a failing test's results
<code>dbt run --select my_model --debug</code>	Verbose log output
<code>dbt run --fail-fast</code>	Stop at first error

Pro tips

- Always compile before running on big projects — it's cheap and catches Jinja errors.
- For models with stacked views/ephemerals, an SQL error can appear in a downstream model that's "really" caused upstream. Temporarily materialize the chain as `table` to surface the error where it originated.
- Keep `dbt.log` open in a separate terminal pane during debugging — `tail -f`.
- `dbt show` is the fastest way to see what a failing test query actually returns.

Key takeaways

- Four error types map to four phases of execution — error type tells you where to look.
- `target/compiled/` and `target/run/` are your friends — the SQL `dbt` actually generated.
- `dbt.log`, `--debug`, and `dbt debug` cover most setup issues.
- For database errors, copy the SQL out of `target/run/` and run it manually.
- `dbt show` lets you peek at the rows behind any test or model.

Related

- `dbt compile` (inspect generated SQL).
- `dbt show` (preview results).
- `dbt parse` (fast parse-only check).
- Global configs / `--debug`, `--fail-fast`.

```
Compiled inline node is:
select * from "jaffle_shop"."main"."raw_orders"
```

Both forms also write to `target/` in addition to printing.

Performance/connection flags

--no-populate-cache (dbt-level flag)

Skips initial cache population of warehouse metadata. If metadata is later needed, `dbt` incurs a cache miss and queries the warehouse on demand.

```
dbt --no-populate-cache compile --select my_model
```

--no-introspect (compile-level flag)

Disables introspective queries. If a resource definition requires running one (e.g., to expand `*` to column names), `dbt` raises an error rather than connecting.

```
dbt compile --no-introspect
```

 When models use introspective queries (`run_query`, `get_columns_in_relation`, etc.), compiled SQL depends on warehouse metadata. Compilation may be incomplete or inconsistent depending on the warehouse state at compile time.

Where the SQL lives

Folder	Contains
<code>target/compiled/<project>/<path>/<model>.sql</code>	The bare compiled <code>select</code>
<code>target/run/<project>/<path>/<model>.sql</code>	The compiled SQL wrapped in materialization DDL (create table as, merge into, etc.)

Use the **compiled** version for ad-hoc query/debug; the **run** version to see exactly what `dbt` would send.

Common patterns

```
# Compile everything
dbt compile

# Just one model
dbt compile --select stg_orders

# A test
dbt compile --select unique_stg_orders_order_id

# Ad-hoc Jinja
dbt compile --inline "select count(*) from {{ ref('stg_orders') }} where
status = 'returned'"

# Compile without hitting the warehouse for metadata
dbt --no-populate-cache compile --no-introspect

# Compile snapshots (v1.12+) – generates SQL with dbt_valid_from/to logic
dbt compile --select my_snapshot
```

Behavior Changes (Flags)

Source: <https://docs.getdbt.com/reference/global-configs/behavior-changes> Type: Documentation · Checkpoint 2

What they are

A specific category of dbt flags that **opt projects into new runtime behaviors gradually**. They give a migration window between an old behavior and a new, better one. Set in the `flags:` dictionary in `dbt_project.yml`.
In **dbt Fusion**, all behavior-change flags are removed — the new behavior is always on. Read this doc primarily for Core 1.x and dbt Latest release track migrations.

Why they exist

- **Better defaults for new users** without breaking existing projects overnight.
- **Migration window for existing projects** — opt in when ready, see warnings until you do.
- **Maintainability for dbt itself** — every branching behavior is testing/cognitive overhead; flags are meant to be temporary.

The lifecycle (three phases)

1. **Introduction** — new behavior gated behind flag, **disabled by default**. Old behavior preserved.
2. **Maturity** — default flips to `true`. New behavior is the default. You can still opt out (`false`) to keep old behavior, but you'll see deprecation warnings.
3. **Removal** — flag and old behavior deleted from dbt entirely. Significant advance notice given.

What counts as a “behavior change”

Same project code + same command → different result.
Examples: - dbt newly raises a validation **error** (not a warning). - Built-in macro signature changes (your custom override may break). - Adapter renames/removes a method on `{{ adapter }}`. - Breaking change to artifact JSON contract (removed required field, etc.). - Removed structured-log field.
Not behavior changes: - Bug fixes (the prior behavior was defective). - New **warnings** (not errors). - Updated human-readable log strings. - Non-breaking artifact additions.

Where to set them

`dbt_project.yml`, under `flags:`. They're project-code-adjacent, so they belong in version control with PR review.

```
flags:
  require_explicit_package_overrides_for_builtin_materializations: true

  require_resource_names_without_spaces: true
  source_freshness_run_project_hooks: true
  skip_nodes_if_on_run_start_fails: false
  state_modified_compare_more_unrendered_values: false
  validate_macro_args: false
  require_generic_test_arguments_property: true
# ... many more
```

To opt out of a matured flag (keep old behavior): set its value to `false`. You'll see warnings — either fix the underlying issue (flag → `true`) or suppress via `warn_error_options.silence`.

Compile vs run vs build vs parse

Command	Compiles SQL	Connects to warehouse	Executes SQL
dbt parse	✗ (syntax check only)	✗	✗
dbt compile	✓	usually (for cache/introspection)	✗
dbt run	✓	✓	✓ (models only)
dbt build	✓	✓	✓ (everything)

Key takeaways

- `dbt compile` renders compiled SQL into `target/` without executing.
- Two ways to inspect in-terminal: `--select <node>` and `--inline "<sql>"`.
- Not a prerequisite for `run/build/test` — they compile internally.
- For pure project parsing without warehouse access, use `dbt parse`.
- For metadata-free compile, combine `--no-populate-cache` and `--no-introspect`.
- v1.12+ extends compile to snapshots.

Related

- `dbt parse` — parse-only validation.
- `dbt show` — compile + execute + preview rows.
- Debugging errors guide — the consumer of compile output.

Examples of common behavior-change flags

Flag	Effect when true
<code>require_explicit_package_overrides_for_builtin_materializations</code>	Packages can't silently override <code>table/view/incremental</code> etc.
<code>require_resource_names_without_spaces</code>	Spaces in model names raise an error
<code>source_freshness_run_project_hooks</code>	<code>dbt source freshness</code> runs on-run-start/-end hooks
<code>skip_nodes_if_on_run_start_fails</code>	Failed on-run-start hook skips selected resources
<code>state_modified_compare_more_unrendered_values</code>	Compare unrendered values during <code>state:modified</code> — reduces false positives across envs
<code>require_yaml_configuration_for_mf_time_spines</code>	MetricFlow time spine must be in YAML, not a SQL config block
<code>require_batched_execution_for_custom_microbatch_strategy</code>	Custom microbatch macros run in batches
<code>validate_macro_args</code>	dbt validates macro call arguments against declarations
<code>require_generic_test_arguments_property</code>	Generic tests must use arguments instead of inline kwargs
<code>enable_truthy_nulls_equals_macro</code>	Null-safe equality macro behavior

Adapter-specific flags

Adapters (Databricks, Redshift, BigQuery, Snowflake) ship their own behavior-change flags too — for things like `use_info_schema_for_columns` (Databricks), `redshift_skip_autocommit_transaction_statements`, `bigquery_use_batch_source_freshness`, `snowflake_default_transient_dynamic_tables`. Same lifecycle pattern; same `flags:` block.

Practical migration playbook

1. After upgrading, look for deprecation warnings in your run logs.
2. For each flag with TBD maturity: assess whether you can adopt now.
3. To adopt: fix the underlying issue, set the flag to `true`, confirm builds pass.
4. To defer: set explicitly to `false` so the warning doesn't surprise you when the default flips later.
5. Track deprecation/email notifications — they're sent before maturity dates.

Key takeaways

- Behavior-change flags = controlled migration mechanism for breaking changes.
- Three phases: introduction → maturity → removal.
- Configure in `flags`: block of `dbt_project.yml`.
- Default `true` = the new behavior is on; set `false` to revert (you'll see warnings).
- **Fusion removes them entirely** — new behavior always wins.

About Flags (Global Configs)

Source: <https://docs.getdbt.com/reference/global-configs/about-global-configs> **Type:** Documentation · Checkpoint 2

What they are

"Flags" (a.k.a. global configs) control **how** dbt runs — log levels, fail-fast behavior, partial parsing, color output, caching, defer, threads, target, etc. They differ from **resource configs** (which describe **what** to build).

Where you can set them

Three layers. Most flags can be set in more than one; CLI > env var > project.

1. **CLI option** at invocation time (`--full-refresh`, `--debug`).
2. **Environment variable** — `DBT_F00` (v1.10 and earlier) or `DBT_ENGINE_F00` (v1.11+).
3. **dbt_project.yml flags**: **block**.

```
# dbt_project.yml
flags:
  fail_fast: true
  partial_parse: true
  use_colors: true
  warn_error_options:
    include: all
    silence: ["NoNodesForSelectionCriteria"]
```

Some flags can **only** live in `dbt_project.yml`. Some can **only** be set per-invocation. The reference table in this doc lists each flag's allowed locations.

Accessing flags from Jinja

The `flags` context variable exposes flag values inside templates:

```
on-run-start:
  - '{{ log("I will stop at the first sign of trouble", info=true) if
    flags.FAIL_FAST }}'
```

⚠️ Don't use `flags` to drive `ref/source/configurations` that dbt resolves at parse time — flags can vary per invocation, parse-time resolution can't.

Commonly used flags (quick reference)

Flag	Type	What it does
target (<code>--target dev</code>)	string	Which target/profile to use
profile	string	Which top-level profile in <code>profiles.yml</code>
debug (<code>--debug</code>)	bool	Verbose logging
fail_fast (<code>-x</code>)	bool	Stop on first error
full_refresh (<code>--full-refresh</code>)	bool	Force-rebuild incremental models
empty (<code>--empty</code>)	bool	Build schemas with no rows
sample (<code>--sample</code>)	string	Time-bounded sample data
partial_parse	bool (default true)	Reuse parser state across runs for speed
populate_cache	bool (default true)	Warm warehouse metadata cache
introspect	bool (default true)	Allow introspective queries during compile
defer / state / favor_state	bool/path	Slim CI defer pattern
warn_error / warn_error_options	bool/dict	Treat warnings as errors
store_failures	bool	Persist failing test rows to a table
printer_width	int	Console width for output
use_colors	bool	Colored output
write_json	bool	Write <code>manifest.json</code> etc.
log_level / log_format / log_path	enum/path	Log shaping
quiet	bool	Suppress non-error logs
resource-type / --exclude-resource-type	string	Filter what runs
event_time_start / event_time_end	datetime	Microbatch backfill window
cache_selected_only	bool	Cache only nodes you selected

Related

- Deprecations index (`/reference/deprecations`).
- About global configs (the broader flag system).
- `warn_error_options` (silencing matured-but-opted-out warnings).
- Upgrading guides per dbt release.

CLI vs project — when each makes sense

- **CLI**: per-invocation overrides — `--target prod`, `--full-refresh`, `-x` to fail fast.
- **Env var**: CI/CD secrets (`DBT_PROFILES_DIR`) and consistent invocation context (`DBT_TARGET=prod`).
- **dbt_project.yml**: durable defaults the team needs every time (`partial_parse: true`, `printer_width: 120`).

Targets (`--target`)

Run the same code against different environments:

```
dbt run --target dev
dbt run --target prod
dbt build --target staging
```

Targets live in `profiles.yml`. Without `--target`, dbt uses the profile's default target.

The v1.11+ env-var rename

Starting v1.11, env vars are prefixed `DBT_ENGINE_` instead of `DBT_` (e.g., `DBT_ENGINE_FAIL_FAST` vs `DBT_FAIL_FAST`). Functional behavior is identical. v1.10 and earlier still use `DBT_`.

Behavior change flags (related but distinct)

There's a separate category of flags called **behavior changes** that gate migration between old and new dbt behaviors. They have a strict lifecycle (intro → maturity → removal) and live in the same `flags`: block. See *Behavior changes* cheat sheet.

High-impact recipes

```
# Slim CI
dbt build --select state:modified+ --defer --state path/to/prod

# Diagnose a flaky build with full logs
dbt --debug build --log-level debug --fail-fast

# CI-friendly: warnings become errors
dbt build --warn-error

# Zero-cost compile validation
dbt build --empty

# Store failing test rows for inspection
dbt test --store-failures
```

Key takeaways

- Flags = how dbt runs; resource configs = what dbt builds.
- Set in three places: CLI, env var, `dbt_project.yml`. CLI overrides env overrides project.
- Use `flags` Jinja var to read values in macros/hooks (but not at parse-time-resolved configs).
- Each flag's reference page lists where it can be set, env-var name, and CLI form.
- v1.11+ rename: `DBT_F00` → `DBT_ENGINE_F00`.

Related

- Behavior changes (cousin doc — migration-controlling flags).
- Environment variable configs.
- Slim CI pattern (uses `defer/state/favor_state`).
- `warn_error_options` (granular warning-to-error control).